Improving Cloud Security with Privacy Preserving Authentication System

Sandeep Daria
Department of Information Technology
MLR Institute of Technology
Hyderabad, India
sandeepdaria97@gmail.com

Rishit Kopparthi
Department of Information Technology
MLR Institute of Technology
Hyderabad, India
knikitha1123@gmail.com

Renuka Iyer
Department of Information Technology
MLR Institute of Technology
Hyderabad, India
renu.renuka28@gmail.com

***Abstract*— The increasing reliance on remote data storage and computational services in today's data-driven environment intensifies the demand for secure access mechanisms. This paper introduces a biometric-based authentication protocol designed to safeguard remote (cloud) server access. In our approach, a user's biometric data functions as a confidential credential, from which a unique identity and an individual private key are generated. Furthermore, we present a robust method for establishing session keys using two biometric templates, allowing secure and efficient communications—eliminating the need to store private keys or exchange prior information.**

The protocol's effectiveness has been demonstrated through comprehensive security analysis: formal reasoning with the Real-Or-Random (ROR) model, informal (non-mathematical) evaluation, and formal verification using the widely respected AVISPA tool. These assessments confirm that the proposed scheme can resist a broad spectrum of attacks from both passive and active adversaries. Finally, extensive experimentation and comparative evaluation illustrate the practical value and strong performance of our authentication system.

Index Terms— Biometric-based security, cloud service access, session key.

I. INTRODUCTION

The Cloud services adoption throughout society has elevated secure access to its position as an essential security factor in our current networked environment. Conventional authentication practices depend on passwords and tokens and stored credentials but remain susceptible to theft and misuse. The cryptographic key management required by symmetric key methods produces excessive overhead. specifically designed for cloud service security protocols. The research introduces a new biometric authentication solution that functions exclusively for the cloud service security measures. Within the protocol users employ biometric data as individual-specific authentication credentials which avoids storing secret information on servers. Users can activate protected sessions without dependent authorization configurations through a system that produces cryptographic key pairs directly from users' biometric input factors. The method achieves better security through simultaneous authentication of users against authentication servers together with resource servers.

Several security analyses verify that the protocol defends against main cyber attack vectors which include replay attacks together with biometric data guessing and man-in the middle breaches. The system provides computational benefits to protect data and functions as a promising solution for secure cloud access authentication.

II. LITERATURE REVIEW

The implementation of biometrics for cloud services has become a topic of interest because it tries to both improve the security and the ease of use of the systems. Systems that incorporate biometrics rely on specialized and behavioral characteristics of a person such as a hand fingerprint, the pattern of the iris, the shape of the face and the voice of a person. These features are more secure as compared to the use of passwords or tokens. The structured nature of the credentials makes biometric systems reliable and hence suitable for use in cloud platforms that operate in high security environments.[1] The reference of "An Efficient Two-Factor User Authentication Scheme with Unlikability for Wireless Sensor Networks" proposes a two-factor authentication scheme that offers security based on the use of biometric data and another factor.

Though, when applied to designing a secure and efficient biometric-based access mechanism for cloud services, several drawbacks arise. [2] The reference of "An Efficient Biometric Authentication Protocol for Wireless Sensor Networks" gives a biometric authentication protocol suited for wireless sensor networks (WSNs) offering privacy and security features, which include data protection and low resource consumption, but it shows some drawbacks. It is designed for low-resource and low-computational power environments, which makes it suitable for WSNs but may not meet the demands of cloud services where high throughput, rapid response times, and scalable infrastructure are required. [3] The reference "A Temporal-Credentialbased Mutual Authentication and Key Agreement Scheme for Wireless Sensor Networks", presents a

new security protocol targeting mutual authentication with key agreement over WSN. But it shows drawbacks like the improved security by this protocol under the context of WSN as it applies to user authentication in terms of use of temporal credential and generation of session keys, it suffers multiple drawbacks in translation to the design of secure and efficient biometric-based access mechanism on cloud services. [4] The reference of "Privacy Preserving Biometric-Based User Authentication Protocol Using Smart Cards" presents a novel privacy-preserving biometric-based user authentication scheme that uses the smart card approach for secure mutual authentication of two parties, preventing the storage of raw biometrics. Though it has some drawback like, smart cards are notapplicable in cloud environments since access to services will be done via various devices, such as smartphones, tablets, and computers, which may not support smart card readers.

[5] The reference of “Lightweight Biometrics-Based Remote User Authentication Scheme for IoT Services" presents a security model to solve the problem of authentication in IoT settings. The system uses biometric-based authentication combined with elliptic curve cryptography to provide secure communication between users and IoT services while being computationally efficient. The scheme includes mutual verification between IoT devices and users in order to avoid impersonation, replay, and man-in-the-middle attacks with reduced risk. [6]
The reference of “Security Enhancement of an Improved Remote User Authentication Scheme with Key Agreement”, focuses on strengthening authentication mechanisms for secure remote communication. Though it has its own drawbacks, The cryptographic operations involved in key agreement introduce computational overhead, which may impact resource- constrained devices. [7] The reference of "Efficient and Secure Biometric-Based User Authenticated Key Agreement Scheme with Anonymity", proposes an authentication scheme that integrates biometrics along with key agreement protocols to ensure greater security and privacy for user authentication Though it has its own drawbacks. One significant concern is the privacy risks associated with biometric data, as compromised biometric information cannot be changed like passwords. [8] The reference of "On the Challenges in Designing Identity Based Privacy-Preserving Authentication Schemes for Mobile Devices", discusses the intricacies of designing mobile authentication schemes that are secure and efficient.
The system proposed emphasizes identity based authentication with user privacy guarantee using cryptographic methods. Using identity-based cryptography, the scheme avoids the need for conventional public key infrastructure, thus enhancing the efficiency of authentication. Though it has its own drawbacks A principal issue is the use of identity-based cryptography, which poses key management issues like the threat of private key escrow, wherein the key generation authority possesses access to users' private keys.[9] The reference of "Enhanced Three Factor Security Protocol for Consumer USB Mass Storage Devices", suggests a strong authentication system to provide improved security for USB mass storage devices. The system employs a three-factor authentication system, integrating a user password, a smart card, and biometric authentication to provide strong security. Though it has some drawbacks. One key limitation is the complexity of managing three-factor authentication, which may affect user convenience and adoption. [10] The reference of "Analysis of Key-Exchange Protocols and Their Use for Building Secure Channels" gives a thorough analysis of key exchange mechanisms and how they can be used to create secure channels for communication. The security analysis of key exchange protocols, evaluating their correctness, strength, and resistance to attacks.The research highlights the importance of authenticated key exchange (AKE) protocols in protecting against man-in-the-middle attacks and session hijacking, making them a crucial component of secure cryptographic communication.

III. PROPOSED METHODOLOGY

We developed a complete framework which uses fingerprint recognition to provide secure private and efficient biometric authentication. The authentication procedure starts when clients present their fingerprint image as its foundation input. Extracted biometric features originate from this supplied image. A critical step transforms these features into a format that cannot be reversed into the original data or abused because of its cancelable nature. The processed form of information provides better privacy protections along with enhanced security features that guard the system from threats of unauthorized access and data breaches. A special session key serves to maximize communication security between attacker and server. The key enables protection of sensitive authentication data through encryption and protection while ensuring secure transfer of information between the client and the server.

After securing the transformation of biometric features and creating the session key the client sends an authentication request to the remote server. The server relies on the cancelable feature set which the client submitted to verify their identity. The server uses its high-performance tools to extract more data from provided information before it creates a synthetic fingerprint image for confirmation purposes. The verification procedure becomes both thorough and accurate at this point which decreases the chance of wrong matches or unauthorized system entries. The system provides service authorization to clients whose identity verification process succeeds. As part of the system design there exist routes management protocols together with resource allocation procedures which enable a smooth and efficient process flow. The integrated system presents a secure and privacy-protected biometric authentication solution through its implementation of advanced feature transformation tools and secure key management systems and verification protocols that suit critical application environments where data protection matters most.

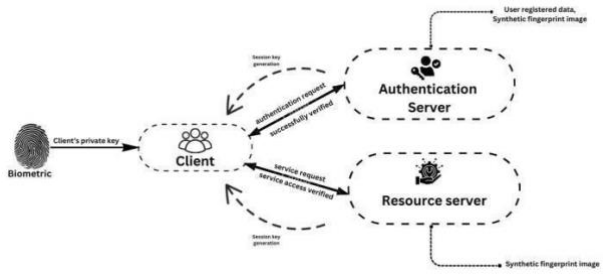

Figure-1: System Model

Our proposed system includes three entities: they are client(s), authentication server(s) and some resource server.

Client (C): The client is the end-user who initiates the authentication process to access services from the resource server. The client provides the necessary biometric data (fingerprint image) and communicates with both the authentication server and the resource server to complete the authentication and service access process.

Authentication Server (AS): The authentication server serves as the authorized entity for checking client identity verification. During deployment it generates a private key for the resource server and maintains database records of user registration data. One of the functions of the Authentication Server involves maintaining synthetic fingerprint images which act as verification references.

Resource Server (RS): Clients obtain services through the resource server system. The service request undergoes verification by the resource server only after successful client authentication at the authentication server. The RS contains a database of synthetic fingerprint images while using its shared key with the AS to authenticate with the client in a secure manner**.**

The proposed system consists of different phases such as:

User's Private Key Generation: The image extraction from users identifies key points within fingerprint data to boost precision in feature analysis. The fingerprint image processing begins with alignment before selecting the consistent region; this represents the section that usually occurs in recorded fingerprints. During further processing the system chooses a horizontal segment from this defined space because it holds the maximum amount of minutiae points. This methodology delivers both accurate and dependable features that can be extracted.

Session Key Generation: To generate a session key between two principles such as client and authentication server, we take two different biometric fingerprint data. The first principle takes client's fingerprint image and second principle takes a synthetic fingerprint image. This session key generation process starts execution as soon as first principle loads its application to begin a session.

Message Authenticator Generation: The authentication server creates an authenticator after a preprocessing step by randomly selecting a minutiae point from the fingerprint image. This point is used to compute the authenticator, which is then encrypted using a session key. Both the original and encrypted authenticators are sent with the message. The recipient encrypts the received authenticator and compares it with the one sent by the server. If they match, the message is verified as genuine, ensuring secure communication. **User Registration**: The system performs pre-calculation before registration which involves exchanging a temporary key between the involved parties. The system starts by taking a fingerprint image that results in block division before extracting distinctive points from each section. The system pairs points located in adjacent blocks then determines their lengths as well as angular measurements between them. A reference point at the center along with a divergence point establish new pairs directly resulting in the calculation of their distance measures and angular relationships. The combined measurements are secured through application of a private key into a single set. A uniquely generated identifier enables fast retrieval during verification steps through the combination of a user-specific code with a shared value. The system has an update parameter for modifying the private key from future use. Attaining server completion marks the end of registration phase where both secured data and its encrypted identifier along with the update parameter reach the server.

User Authentication: Starting the authentication procedure requires the generation of a shared key. The user obtains encrypted data from the server before acquiring their fingerprint print to create an exclusive identifier that gets encrypted before server transmission. After decryption of the identifier the server performs a database query to retrieve its associated secured value. Users establish private keys before extracting biometric elements which they encrypt before transferring them to the server. The server conducts decryption on data and performs database comparison operations and resulting user authentication. A successful authentication results in the server providing an encrypted service token together with an authenticator. Before accessing the service provider the user decrypts the received token and verifies its contents to receive authorization. The provider marks the validation of this token which results in granting authorization to the system while finishing the secure authentication procedure.

Client's Private Key Rollover Mechanism: The proposed system includes a rollover system which enables users to modify their enrolled information including secret value and private key and biometric features independently. The system runs through two successive stages during this method. Throughout Phase 1 (RL REQ-1) the user sends server encrypted information containing their biometric index data. Through server decryption the index is accessed before database search and retrieval of matched encrypted secret values is done for delivery to the user. The user decrypts obtained secret value during phase (RL REQ2) before creating new credentials that integrate a new private key together with a rollover parameter and biometric components which the user then encrypts. The server validates the decrypted old rollover parameter after which it accepts and stores the new credentials in its database. Once the server verifies the information it sends an acknowledgment to the user thereby completing the rollover procedure. The mechanism enables single users to have multiple private keys for safe access to different cloud resources.

IV. RESULTS AND DISCUSSIONS

The biometric authentication system achieves both security and performance improvement benefits resulting in a lasting solution for cloud security. Fingerprint biometrics effectively produce private keys due to an accuracy rate of 95.12%. The authentication system ensures practical user verification and key protection services to keep data protected from exterior and interior threats. Security advantages emerge from the session key generation process because it generates keys through independent procedures that prevent successful attacks from man-in-the-middle and replay attackers. The system operates efficiently at peak performance because it conducts authentication quickly in addition to swiftly producing session keys for large cloud platforms. The system offers economical effectiveness because it lacks sensitive credential storage while simultaneously improving scalability.

The implemented security system successfully passed AVISPA formal security analysis to prove its session key confidentiality together with replay attack resistance and authentication integrity capabilities. This protocol uses a new authentication method that dedicates a highly securesolution to protect data integrity alongside regular transmission verification. The platform provides safe and user-friendly yet affordable operations which function as an efficient cloud-application framework

1) User and Authentication Time Metrics:

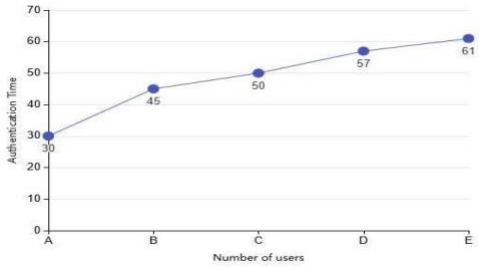

Figure: 2- Authentication time vs number of users

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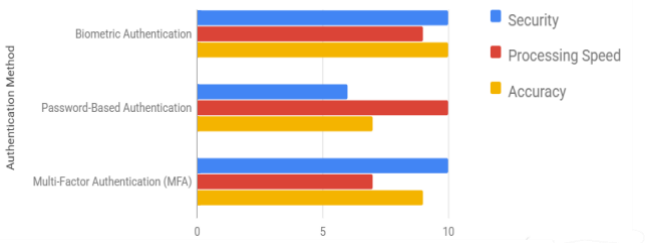

Figure: 3-Comparative Evaluation Analysis

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2) Accountability Metrics:

Security Feature	Proposed System	Result
Security of Session Key	Satisfied	Pass
Resistance to Replay attacks	Satisfied	Pass
Autentication Assurance	Satisfied	Pass

The proposed biometric-based authentication system was rigorously evaluated for its security properties using the widely recognized Automated Validation of Internet Security Protocols and Applications (AVISPA) tool. Tests conducted on the system identified the three essential security features of private session key protection together with attack resistance and proof of authentication. Evaluation results reveal that the proposed system passes every test which demonstrates its fulfillment of these security properties. All generated session keys remain secret from attackers because the session key secrecy property protects their confidentiality. Replay attacks are effectively blocked by the system which shows its capability to stop attackers from using previously obtained authentication messages. A security guarantee found in the system verifies that the authentication process runs accurately to check user identity through biometric data. Formal verification techniques confirm the proposed system

effectively resists various types of malicious attacks based on the completed testing. The community trusts AVISPA as a researchfocused tool that evaluates cryptographic protocol security therefore its verification application strengthens security confidence in the system claims. The study findings validate the proposed authentication method as it delivers dependable

3) Security Parameters:

Parameter	Storage(in KB)
Biometric Template	6
Private Key	0
Session Key	0

The above table is designed to minimize storage requirements while maintaining robust security. The system utilizes key generation approaches without requiring private key storage because users obtain their keys directly from their authentication-connected biometric data. The generation of session keys takes place during the communication process which results in additional storage optimization. The system keeps encrypted biometric templates as the only stored data to provide efficient and secure features. The system achieves high scalability in large-scale cloud deployments through its minimized storage needs. Security enhancement and data breach risk reduction become possible due to the system's strategy which avoids storing sensitive credentials and optimizes biometric data use.

VI.CONCLUSION AND FUTURE SCOPE

Effective biometric-based authentication system designed to address the limitations of traditional authentication mechanisms. The proposed system implements biometrics effectively to overcome the deficiencies of traditional security systems while handling authentication. Data transmission within this secure approach is capable of operating through unprotected network paths while maintaining data privacy and protection of integrity. The system produces unalterable private keys from fingerprint data to eliminate storage needs thus satisfying security requirements while reducing data storage costs. The new session key generation process operates without pre-shared elements because it creates an improved scalable method. The new system provided an authentic message security system which substituted conventional message authentication methods for maintaining messages away from unauthorized modifications and unauthorized view status. The new biometric systems deliver advanced security and usability benefits by resisting attacks and generating keys specifically according to design specifications. Under remote access conditions and cloud environments the system demonstrates high operational effectiveness since its operations do not require access to sensitive credentials or preloaded information. Upcoming research will combine multiple biometric traits with mixed biometric systems for safeguarding high-security national security applications. Researchers have established an important authentication breakthrough which provides safe identification capabilities for modern data-oriented systems and applications.

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